| | |
|---|---|
| | |

⇒ Bus Data — ✕ Later

⇒ Line Data ←

| | St. Bus | End Bus | r | x | b (b/2) |
|---|---|---|---|---|---|
| Line 1 | 1 | 2 | $r_{12}$ | $x_{12}$ | |
| Line 2 | 1 | 4 | $r_{14}$ | $x_{14}$ | |
| line 3 | 2 | 3 | $r_{23}$ | $x_{23}$ | |
| line 4 | 3 | 4 | $r_{34}$ | $x_{34}$ | |

$y_{12}$ $y_{13}$ $y_{23}$ $y_{24}$

⇒ Read Data

$y_{14} = 0$ $n = 4$

```
for i=1, n
  for j=1, n
    if (i=j)
      Y(i,j) = sum of all admittance connected
    else
      Y(i,j) = -y(i,j)
  end
end
```

YBus
100

$n$ Buses

$P_i$, $Q_i$

$$I_i = \sum_{j=1}^{n} Y_{ij} V_j$$

$P_i = P_{geni} - P_{loadi}$ ⇒ real power Injected at bus i

$Q_i = Q_{geni} - Q_{loadi}$ ⇒ reactive power Injected at bus i

If $P_{geni} > P_{loadi} \Rightarrow P_i \Rightarrow$ (+ve) ⟸ Gen Bus

If $P_{geni} < P_{loadi} \Rightarrow P_i \Rightarrow$ (-ve) ⟸ Load Bus

$Q_i \Rightarrow$ +ve / -ve

$I_i \Rightarrow$ relationship with $P_i$ and $Q_i$

$$S_i = V_i I_i^* = P_i + jQ_i$$

$$S_i^* = P_i - jQ_i = V_i^* I_i$$

$$\leftarrow I_i = \frac{P_i - jQ_i}{V_i^*}$$

$$\boxed{\frac{P_i - jQ_i}{V_i^*} = \sum_{j=1}^{n} Y_{ij} V_j}$$

Basic Power flow equations